

# Turbo Equalization for Flexible PAM-2/4 Passive Optical Networks

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**Abstract**—We investigate a turbo equalization (TE) scheme combining soft-input soft-output (SISO) Bahl–Cocke–Jelinek–Raviv (BCJR) detection with low-density parity-check (LDPC) decoding based on the standardized (17280, 14592) LDPC code for passive optical networks. For a two-tap intersymbol interference (ISI) channel, the maximal achievable information rates (AIRs) are analytically derived to quantify the performance gap between separate detection and decoding (SDD) and joint detection and decoding (JDD) under different ISI levels. This analysis reveals the potential gain achievable by JDD and motivates the use of TE as a practical approach to approach JDD performance. Experimental results obtained in a 50-GBd pulse-amplitude modulation (PAM) 2/4 passive optical network (PON) system demonstrate that TE provides received optical power (ROP) gains of 0.33 dB for PAM-2 and 0.56 dB for PAM-4, confirming its effectiveness for bandwidth-limited receivers.

**Index Terms**—Optical communication, turbo equalization, BCJR, LDPC, IM/DD, PON.

## I. INTRODUCTION

TO MEET the rapidly increasing capacity demands in optical access networks, 50G passive optical networks (50G-PON) have been standardized as a key technology for next-generation access systems [1], [2], [3]. In cost-sensitive PON deployments, optical network units (ONUs) are typically equipped with low-cost avalanche photodiodes (APDs), which may result in bandwidth-limited receiver front-ends. When operating at symbol rates as high as 50 GBd, such receivers experience severe ISI due to the combined effects of limited device bandwidth and fiber chromatic dispersion (CD), which significantly degrades receiver sensitivity and constrains the achievable power budget.

To mitigate ISI under bandwidth-limited conditions, various digital signal processing (DSP) techniques have been extensively investigated [3], [4]. At the receiver side,

low-complexity linear and nonlinear equalizers, such as feed-forward equalizers (FFE) and decision feedback equalizers (DFE), are widely adopted in practical PON systems owing to their implementation simplicity and low latency [5]. However, under strong ISI conditions, especially in high-speed PAM systems, FFE causes noise enhancement, while DFE suffers from error propagation, leading to limited bit error rate (BER) performance improvements.

To further enhance ISI mitigation capability, sequence-based detection techniques such as maximum likelihood sequence estimation (MLSE) and maximum a posteriori (MAP) detection have been proposed for bandwidth-limited optical links [6], [7], [8], [9]. These techniques can effectively handle severe ISI, offering significant performance gains over conventional FFE/DFE-based solutions.

In current PON systems, SDD, where equalization and forward error correction (FEC) decoding are performed independently, represents the state-of-the-art solution [3], [4], [10]. While SDD offers a favorable complexity–performance trade-off, it is inherently suboptimal in the presence of strong ISI, since the soft information generated during symbol detection does not fully capture the residual ISI. This limitation becomes increasingly pronounced for higher-order modulation formats. Although PAM-4 is not currently adopted in 50G PON standards, it has attracted significant research interest as a potential option for boosting downstream data rates without increasing the symbol rate [11].

TE, which iteratively exchanges soft information between a SISO equalizer and a channel decoder, provides an effective means to approach JDD performance with manageable complexity [12], [13], [14]. The remainder of this letter is organized as follows. Section II introduces the principle of TE. Section III presents the simulation setup and results, while Section IV details the experimental setup and corresponding results. Finally, Section V concludes the letter.

## II. PRINCIPLE OF TURBO EQUALIZATION

In the SISO TE framework [15], as depicted on the right-hand side of Fig. 1, the BCJR-based equalizer iteratively operates on the received noisy signal and computes a posteriori symbol-wise log-likelihood values ( $L$ -values). These symbol-level likelihood metrics are subsequently transformed into bit-level  $L$ -values through a soft demapping operation based on marginalization, denoted as  $L_{p,bcjr}$ .

In contrast to hard-decision (HD) equalization, which irrevocably selects a single symbol or bit and thereby loses reliability information, the soft-decision (SD) approach retains

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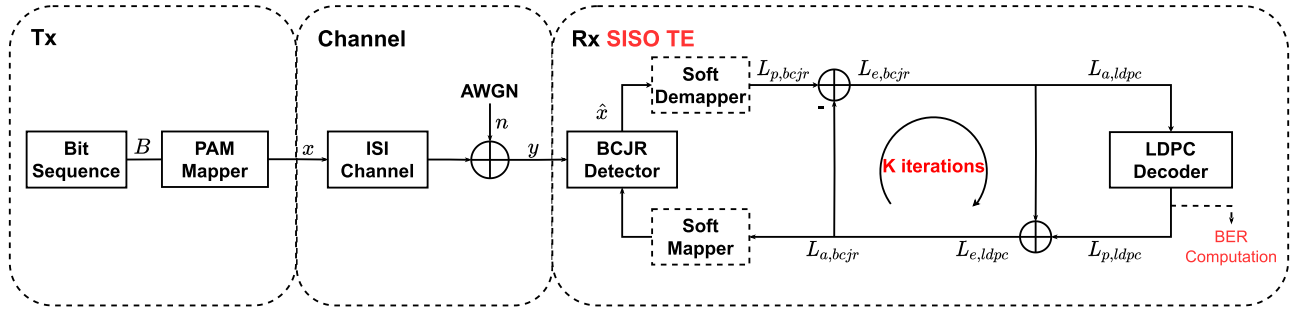


Fig. 1. The simulated transmission system. Dashed blocks indicate the soft mapper/demapper required for PAM-4.

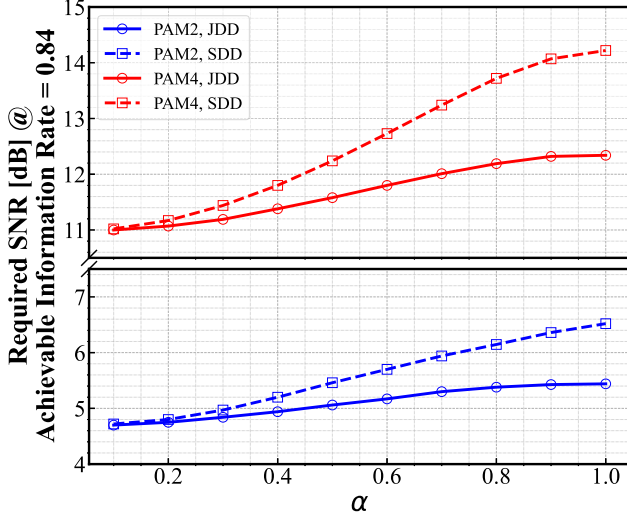


Fig. 2. Required SNR to achieve an information rate of 0.84 (PAM-2) and 1.68 (PAM-4) under different values of  $\alpha$ .

uncertainty information associated with competing hypotheses. By subtracting the a priori information  $L_{a,bcjr}$ , the extrinsic information can be obtained as

$$L_{e,bcjr} = L_{p,bcjr} - L_{a,bcjr}, \quad (1)$$

where, in the initial TE iteration, the a priori information is initialized as  $L_{a,bcjr} = 0$ . The resulting extrinsic information  $L_{e,bcjr}$  is then fed into the LDPC decoder.

The (17280,14592) LDPC code is decoded using the min-sum algorithm [16]. In this work, the baseline SDD performance is defined as BCJR detection followed by LDPC decoding with  $I = 20$  iterations. The LDPC decoder produces extrinsic bit-wise  $L$ -values  $L_{e,LDPC}$ , computed as

$$L_{e,ldpc} = L_{p,ldpc} - L_{a,ldpc}, \quad (2)$$

where  $L_{p,ldpc}$  and  $L_{a,ldpc}$  denote the a posteriori and a priori  $L$ -values, respectively. These extrinsic  $L$ -values serve as a priori information for the next TE iteration and must be converted back to symbol-wise  $L$ -values for the BCJR detector. Ultimately, the iterative loop progressively refines the reliability of the estimated bits, analogous to the iterative decoding principle underlying turbo codes.

### III. SIMULATION SETUP AND RESULTS

Fig. 1 shows the simulated transmission system. In the transmitter (Tx), source bits are mapped to PAM-2 symbols

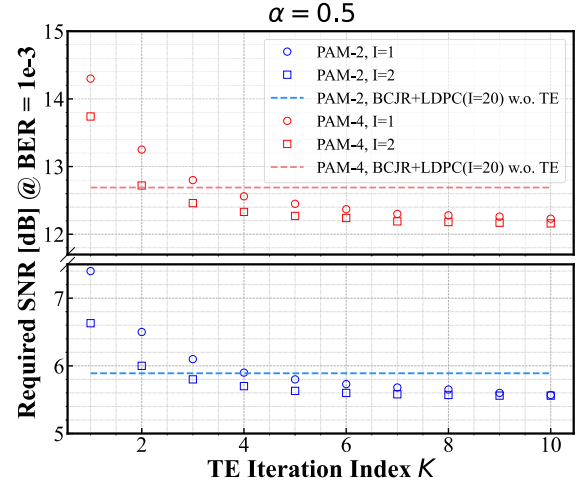


Fig. 3. Required SNR at  $\text{BER} = 10^{-3}$  for the case of  $\alpha = 0.5$  with PAM-2/4.

(e.g., 0, 1) or to PAM-4 symbols using Gray coding, defined as  $0 \mapsto 00, 1 \mapsto 01, 2 \mapsto 11, 3 \mapsto 10$ .

The channel is modeled as a unit-memory ISI link impaired by additive white Gaussian noise (AWGN), which provides a reasonable approximation for bandwidth-limited optical receivers and has been widely adopted in prior work [12], [13]. The received signal  $y_i$  at time index  $i$  is expressed as

$$y_i = x_i + \alpha x_{i-1} + \omega_i, \quad (3)$$

where  $x_i, x_{i-1} \in \mathcal{X}$  denote the transmitted PAM symbols,  $\alpha \in [0, 1]$  controls the ISI level, and  $\omega_i \sim \mathcal{N}(0, \sigma^2)$  denotes AWGN.

This simplified channel model reflects the equivalent channel after FFE and post-filtering (PF), where the ISI is deliberately introduced and known, and the noise is approximately white, allowing the subsequent MAP algorithm to effectively mitigate the ISI. At the receiver (Rx), a SISO TE process [15] is performed. By iteratively exchanging soft information between the BCJR detector and LDPC decoder, we aim at approaching JDD performance.

As shown in Fig. 2, a lower bound on the achievable information rate for the considered channel/receiver model can be estimated by sampling long input-output sequences and performing a forward sum-product recursion on the joint source-channel trellis [17]. For different values of  $\alpha$ , we compute the required signal-to-noise ratio (SNR) to achieve a transmission rate of 0.84, corresponding to the (17280, 14592)

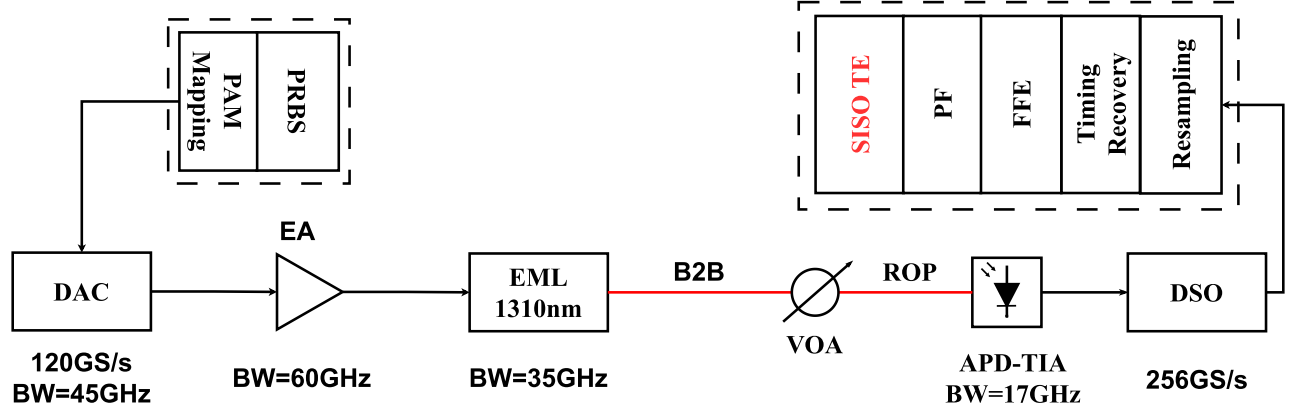


Fig. 4. Experimental setup for high speed PAM-2/4 PON system.

LDPC code in the JDD case. For PAM-4, which carries two bits per symbol, this corresponds to an overall information rate of  $2 \times 0.84 = 1.68$  bits per channel use.

When a BCJR detector is cascaded with the channel, the achievable rate is evaluated using the generalized mutual information (GMI) [18], which quantifies performance under SDD. For a given code rate, the vertical SNR gap between JDD and SDD essentially corresponds to the maximum TE gain when using long channel codes. For example, when  $\alpha = 0.5$ , the gap between the dashed and solid blue curves is 0.38 dB, while the gap between the dashed and solid red curves is 0.6 dB. This indicates that at this ISI level the TE gains for PAM-2 and PAM-4 are up to 0.38 dB and 0.6 dB, respectively.

In Fig. 3, the simulated required SNR to achieve  $\text{BER} = 10^{-3}$  for an ISI level of  $\alpha = 0.5$  is shown. The blue and red dashed curves represent the baseline performance, i.e., a BCJR detector with  $I = 20$  LDPC decoding iterations without TE, reaching 5.9 dB and 12.7 dB for PAM-2 and PAM-4, respectively. The circle and square markers indicate the performance with TE, where circles correspond to the number of LDPC decoding inner iterations  $I = 1$  and squares to  $I = 2$ . As the TE iteration index  $K$  increases from 1 to 10, TE surpasses the baseline at  $K = 4$  for  $I = 1$  and at  $K = 3$  for  $I = 2$ . The final TE gains after  $K = 10$  iterations are 0.34 dB and 0.55 dB for PAM-2 and PAM-4, respectively, consistent with the theoretical TE gains obtained from Fig. 2.

#### IV. EXPERIMENTAL SETUP AND RESULTS

Fig. 4 shows the experimental setup of a flexible PAM-2/4 PON system. A pseudorandom bit sequence (PRBS) is generated and mapped to a PAM-2/4 signal, which is upsampled to 120 GSa/s, filtered by a root-raised cosine filter with a 0.1 roll-off, and output through a 120 GSa/s digital-to-analog converter (DAC, Micram). After amplification by an electrical amplifier (EA), the signal drives a 35 GHz O-band electro-absorption modulated laser (EML). A variable optical attenuator (VOA) controls the received optical power (ROP). The optical signal is detected by an APD with a transimpedance amplifier (TIA), whose combined bandwidth is 17 GHz, and the signal is captured by a 256 GSa/s digital sampling oscilloscope (DSO) for offline DSP. In this work, the sample data length is set to 5 million.

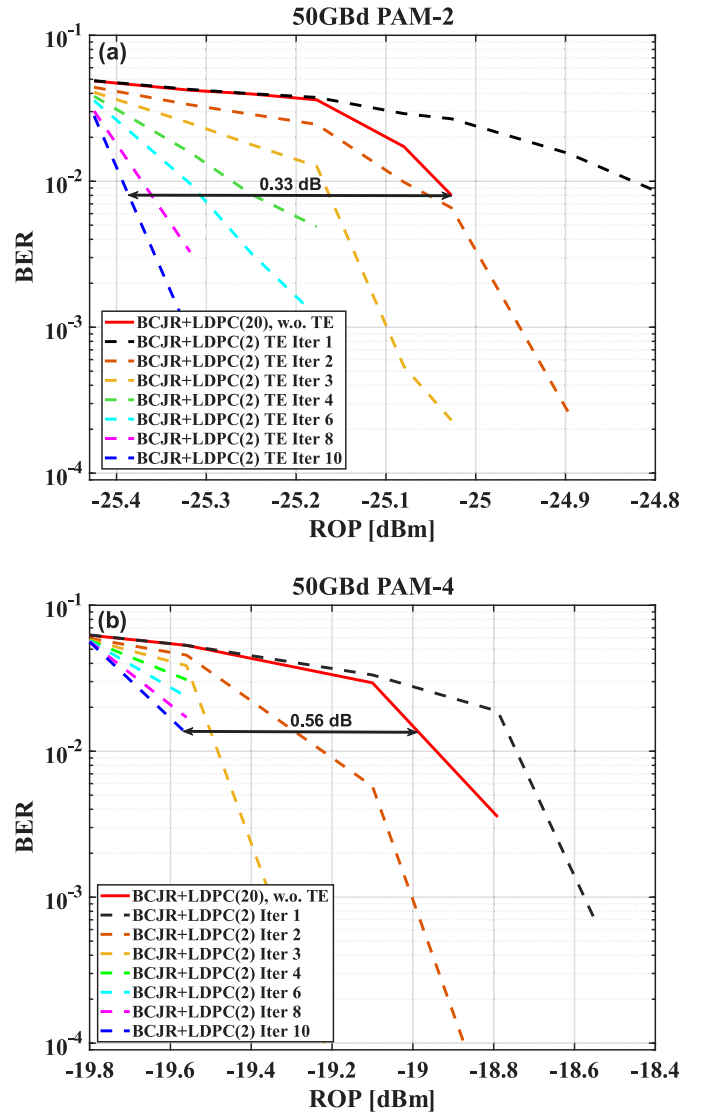


Fig. 5. (a, b) BER vs. ROP for 50 GBd PAM-2 and PAM-4.

In the receiver DSP, the waveform is resampled, timing recovery is performed, and the signal is downsampled to one sample per symbol. A 41-tap linear FFE and a 2-tap noise whitening filter ( $1 + \alpha D$ ) are applied, where  $\alpha \approx 0.5$ .

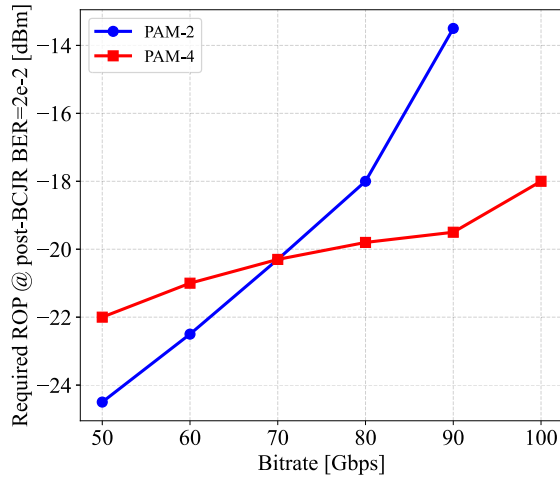


Fig. 6. Required ROP at post-BCJR BER =  $2 \times 10^{-2}$  versus different bitrates for PAM-2 and PAM-4 signals.

Finally, SISO TE is implemented as shown in Fig. 1, using the conventional full-state BCJR algorithm with 4 state transitions for PAM-2 and 16 for PAM-4. This setup enables FEC performance evaluation by using channel adapters to avoid encoding [19].

Figures 5 (a) and (b) present the BER versus ROP characteristics for 50 GBd PAM-2 and PAM-4 signals, respectively. The solid red curve represents the SDD performance (i.e., BCJR+LDPC with 20 decoding iterations), while the dashed curves in various colors depict the TE performance after  $K$  TE iterations, where  $K$  ranges from 1 to 10. In TE, the number of LDPC decoding inner iterations is fixed to 2. After 10 TE iterations, the TE gain is 0.33 dB for the 50GBd PAM-2 case and 0.56 dB for the 50 GBd PAM-4 case. The TE gains in experiments are very close to those obtained by simulations/theory.

Fig. 6 illustrates the required ROP at a post-BCJR BER of  $2 \times 10^{-2}$  for PAM-2/4 systems at different bitrates. PAM-2 offers better receiver sensitivity at lower rates ( $< 70$  Gbps) with much lower ROP than PAM-4, but its performance drops sharply at higher rates due to bandwidth limits. PAM-4, on the other hand, shows a slower ROP increase and outperforms PAM-2 above 70Gbps, making it more suitable for  $\geq 100$ Gbps short-reach optical links with better scalability and power margin.

## V. CONCLUSION

An analytical framework based on AIRs has been presented to estimate the theoretical maximal TE gain for a unit-memory ISI channel, clearly revealing the performance gap between SDD and JDD. Based on this analysis, TE combining a BCJR detector with the standardized (17280, 14592) LDPC code was investigated for a 50 GBd PAM-2/4 PON system.

Both simulation and experimental results confirm that iterative information exchange between the equalizer and decoder effectively mitigates residual ISI in bandwidth-limited

receivers. In experiments employing 25G-class APD receivers, TE achieves received optical power gains of 0.33 dB for PAM-2 and 0.56 dB for PAM-4, in good agreement with theoretical predictions. These results demonstrate that TE is an effective DSP solution to enhance receiver sensitivity in next-generation high-speed PON systems without requiring hardware modifications.

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